



Technical Service Bulletin

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New QST30 Industrial Fan Belt Tensioner

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

This document announces changes to the industrial fan drive options for the QST30 engine. These engine options were originally introduced in Technical Service Bulletin 100650.

The previous design used a torsion spring to provide system tension and a control rod to prevent idler pulley kick back during hard decelerations of the engine. This option change reintroduces a linear spring design in order to increase overall system tension and reduce the possibility of belt walking. A similar design was used on Industrial engines prior to November 2007. The new design removes the torsion spring; however, it maintains the more robust pivot bushing assembly. This pivot assembly has a higher capability than the pivot design used prior to November 2007.

The options affected by this change are listed below. The engine serial number (ESN) first for these options is 37248030, effective 11 February 2011.

- FA5079
- FA5085
- FA5086
- FA5087
- FA5088
- FA5090
- FA5091
- FA5092

The following tables show the parts being removed and added to the options. These parts lists can also be referenced to upfit an engine to the new configuration. If belt walking is experienced with the old design, the engine should be upgraded to the new configuration referenced in the tables below.

Table 1: Option Changes for FA5079				
New Option Part Number	New Quantity	Old Option Part Number	Old Quantity	Part Name
N/A	0	3092289	1	PLATE, CLAMPING
S 608	6	S 608	3	WASHER, LOCK
3092158	1	3092157	1	ASSEMBLY, FAN DRIVE IDLE
169195	2	3202222	2	SCREW, HEXAGON HEAD CAP
4975678	1	4067971	1	BRACKET, BELT TENSIONER
3036900	5	N/A	N/A	PLAIN, WASHER
4068467	1	N/A	N/A	TENSIONER, BELT
S 106	1	N/A	N/A	SCREW, HEXAGON HEAD CAP
68050	1	N/A	N/A	WASHER, PLAIN
S 253	1	N/A	N/A	NUT, REGULAR HEXAGON JAM
146517	1	N/A	N/A	NUT, REGULAR HEXAGON

The option changes for FA5085, FA5086, FA5087, FA5088, FA5090, FA5091, and FA5092 are shown in Tables 2 and 3. Table 2 shows changes that are common to all of the options. Table 3 details the additional changes unique to each option.

Table 2: Common Part Changes for FA5085, FA5086, FA5087, FA5088, FA5090, FA5091, and FA5092					
New Option Part Number	New Quantity	Old Option Part Number		Old Quantity	Part Name
N/A	0	4975435		1	ROD, CONTROL
N/A	0		S 121 B	2	SCREW, HEXAGON HEAD CAP
N/A	0		S 610	2	WASHER, LOCK

Table 2: Common Part Changes for FA5085, FA5086, FA5087, FA5088, FA5090, FA5091, and FA5092

New Option Part Number	New Quantity	Old Option Part Number		Old Quantity	Part Name
N/A	0		S 696	2	WASHER, PLAIN
N/A	0		117267	1	WASHER, COUNTER WEIGHT
N/A	0		3027728	2	SPACER, MOUNTING
N/A	0	4975438		1	BRACKET, BELT TENSIONER
N/A	0		169196	4	SCREW, HEXAGON HEAD CAP
N/A	0		3092289	2	PLATE, CLAMPING
3896083	1	3896083		2	SCREW, HEXAGON HEAD CAP
S 608	6	S 608		3	WASHER, LOCK
3171715	2	3171715		1	SCREW, HEX FLANGE HEAD CAP
4975421	1	4975421		1	ASSEMBLY, IDLER PIVOT
4975611	1		4068607	1	LEVER, IDLER
N/A	0		3002993	2	DOWEL, RING
N/A	0		3012475	2	SCREW, CAPTIVE WASHER CAP
N/A	0		4295351	1	BRACKET, IDLER PULLEY
4068467	1	N/A		0	TENSIONER, BELT

Table 2: Common Part Changes for FA5085, FA5086, FA5087, FA5088, FA5090, FA5091, and FA5092

New Option Part Number	New Quantity	Old Option Part Number	Old Quantity	Part Name
S 106	1	N/A	0	SCREW, HEXAGON HEAD CAP
68050	1	N/A	0	WASHER, PLAIN
S 253	1	N/A	0	NUT, REGULAR HEXAGON JAM
146517	1	N/A	0	NUT, REGULAR HEXAGON
169195	1	N/A	0	SCREW, HEXAGON HEAD CAP

Tables 3, 4, 5, 6, 7, 8, and 9: Unique part changes for each option: FA5085, FA5086, FA5088, FA5090, FA5091, and FA5092 are detailed below.

Table 3: Unique Part Changes for FA5085

New Option Part Number	New Quantity	Old Option Part Number	Old Quantity	Part Name
3092350	3	3092350	4	SCREW, HEXAGON HEAD CAP
3093761	3	3093761	2	SCREW, HEXAGON HEAD CAP
4975612	1	3092060	1	BRACKET, BELT TENSIONER
3092284	1	N/A	N/A	BRACE, IDLER PULLEY
3036900	5	N/A	N/A	WASHER, PLAIN
117267	1	N/A	N/A	WASHER, COUNTER WEIGHT

Table 4: Unique Part Changes for FA5086

New Option Part Number	New Quantity	Old Option Part Number	Old Quantity	Part Name
N/A	0	3281674	1	SCREW, HEXAGON HEAD CAP
3092485	1	3092485	2	SPACER, MOUNTING
4975679	1	4068013	1	BRACKET, IDLER PULLEY
3036900	4	N/A	N/A	WASHER, PLAIN
4295332	1	N/A	N/A	SPACER, MOUNTING
4975677	1	N/A	N/A	BRACE, IDLER PULLEY

Table 5: Unique Part Changes for FA5087

New Option Part Number	New Quantity	Old Option Part Number	Old Quantity	Part Name
3092350	3	3092350	4	SCREW, HEXAGON HEAD CAP
3093761	3	3093761	2	SCREW, HEXAGON HEAD CAP
3171715	2	3171715	1	SCREW, HEX FLANGE HEAD CAP
4975612	1	3092060	1	BRACKET, BELT TENSIONER
4975677	1	N/A	N/A	BRACE, IDLER PULLEY
3036900	4	N/A	N/A	PLAIN, WASHER

Table 6: Unique Part Changes for FA5088

New Option Part Number	New Quantity	Old Option Part Number	Old Quantity	Part Name
3092350	3	3092350	4	SCREW, HEXAGON HEAD CAP
3093761	3	3093761	2	SCREW, HEXAGON HEAD CAP
4975612	1	3092060	1	BRACKET, BELT TENSIONER
3036900	4	N/A	N/A	WASHER, PLAIN
4068050	1	N/A	N/A	SPACER, MOUNTING
3092284	1	N/A	N/A	BRACE, IDLER PULLEY

Table 7: Unique Part Changes for FA5090

New Option Part Number	New Quantity	Old Option Part Number	Old Quantity	Part Name
3092350	3	3092350	4	SCREW, HEXAGON HEAD CAP
3093761	3	3093761	2	SCREW, HEXAGON HEAD CAP
4975612	1	3092060	1	BRACKET, BELT TENSIONER
4975677	1	N/A	N/A	BRACE, IDLER PULLEY
3036900	4	N/A	N/A	WASHER, PLAIN
4295333	1	N/A	N/A	SPACER, MOUNTING

Table 8: Unique Part Changes for FA5091				
New Option Part Number	New Quantity	Old Option Part Number	Old Quantity	Part Name
3092350	3	3092350	4	SCREW, HEXAGON HEAD CAP
3093761	3	3093761	2	SCREW, HEXAGON HEAD CAP
4975612	1	3092060	1	BRACKET, BELT TENSIONER
3036900	4	N/A	N/A	WASHER, PLAIN
4295333	1	N/A	N/A	SPACER, MOUNTING
4975677	1	N/A	N/A	BRACE, IDLER PULLEY

Table 9: Unique Part Changes for FA5092				
New Option Part Number	New Quantity	Old Option Part Number	Old Quantity	Part Name
3171715	2	3171715	1	SCREW, HEXAGON FLANGE HEAD CAP
4975679	1	4295499	1	BRACKET, IDLER PULLEY
3092284	1	N/A	N/A	RACE, IDLER PULLEY

The spacers and washers included in each option as described in Tables 3 through 9 are used to control the system's belt tension. The washers and spacers should be installed as shown in Figure 1 and the top nut tighten until the spacer bottoms out on the bracket. This will make sure the belt tension remains in the desired range.

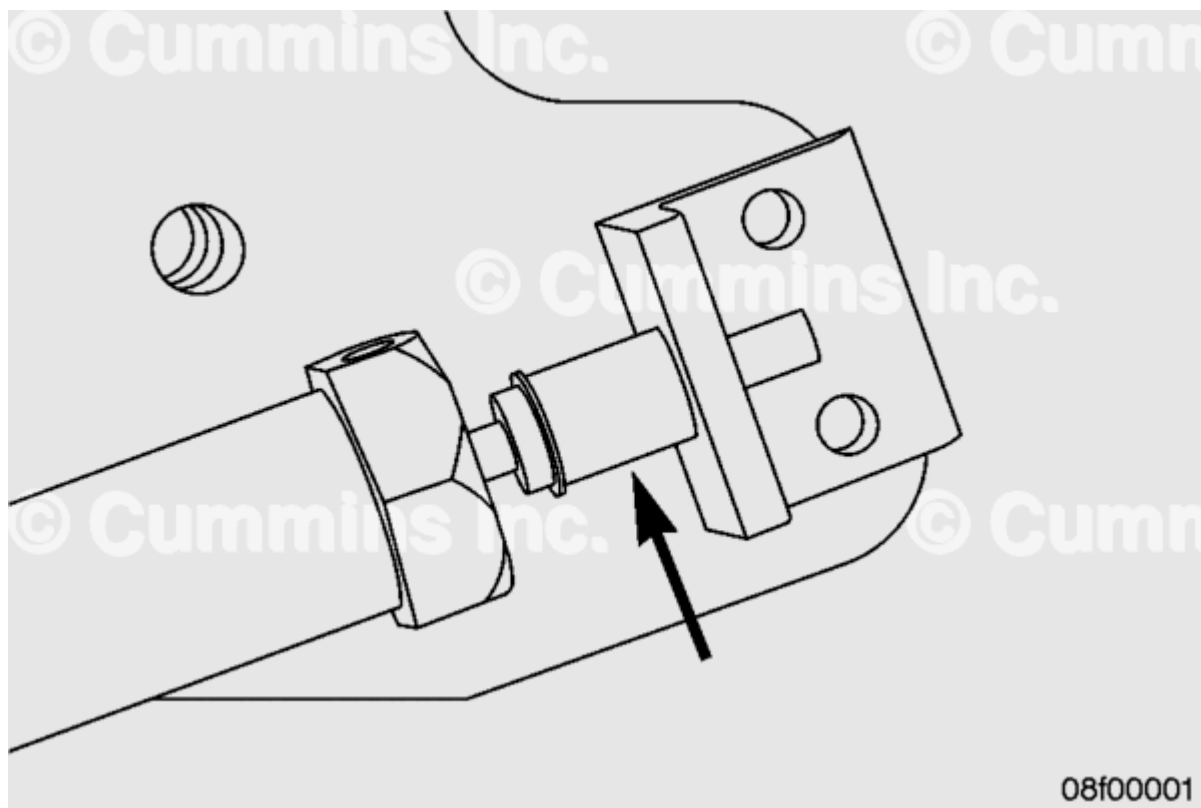


Figure 1

The washers and spacers are installed between the belt tensioner and the idler pulley brace in Figure 1.

Document History

Date	Details
2011-1-6	Module Created
2011-6-15	QSOL Quick Fix Reason: Format Error Notes: Date change and please left justify all text in the tables as indicated. Some is center justified and some is left justified without reason.
2011-6-27	QSOL Quick Fix Reason: QSOL Display Problem Notes: TSB revision did not appear on QSOL.

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